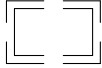
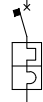
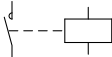
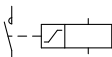
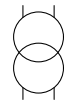


1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																					
SAFETY CRITICAL ELEMENT LIST																					
IFI	2	27/02/2026	AS-BUILT				ADF	GDP	IC												
IFI	1	24/03/2025	GENEERAL REVIEW				ADF	GDP	IC												
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC												
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.												
	Rev. Index	Date	Description				Prepared	Checked	Approved												
Client							Project Title														
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT														
Contractor							Provision of a m-LNG plant for production for ManaEnergy														
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																					
Engineering																					
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																					
Facility Name			Location				Scale		Sheet of Sheets												
TENDRARA			MOROCCO				A4		1 of 17												
Document Title					Doc. Code																
P-PN-OS311.1					ELD01-ING-FR01																
DEHYDRATATION PRECOOLING					Vendor Doc. code																
					IE24090_SE26_P-PN-OS311.1																
			Project : P–PN–OS311.1				Customer : ManaEnergy Managem Group				Object :										
General Contractor :		Manufacture : esa esaelettromeccanica.com		DEHYDRATATION PRECOOLING				Contract : IE24090				Rev. 2		Data : 27/02/2026				Date : 26/03/2024			
								S/N : IE24090_SE26				Rev. 1		Data : 24/03/2025				Sheet : 1		of : 17	

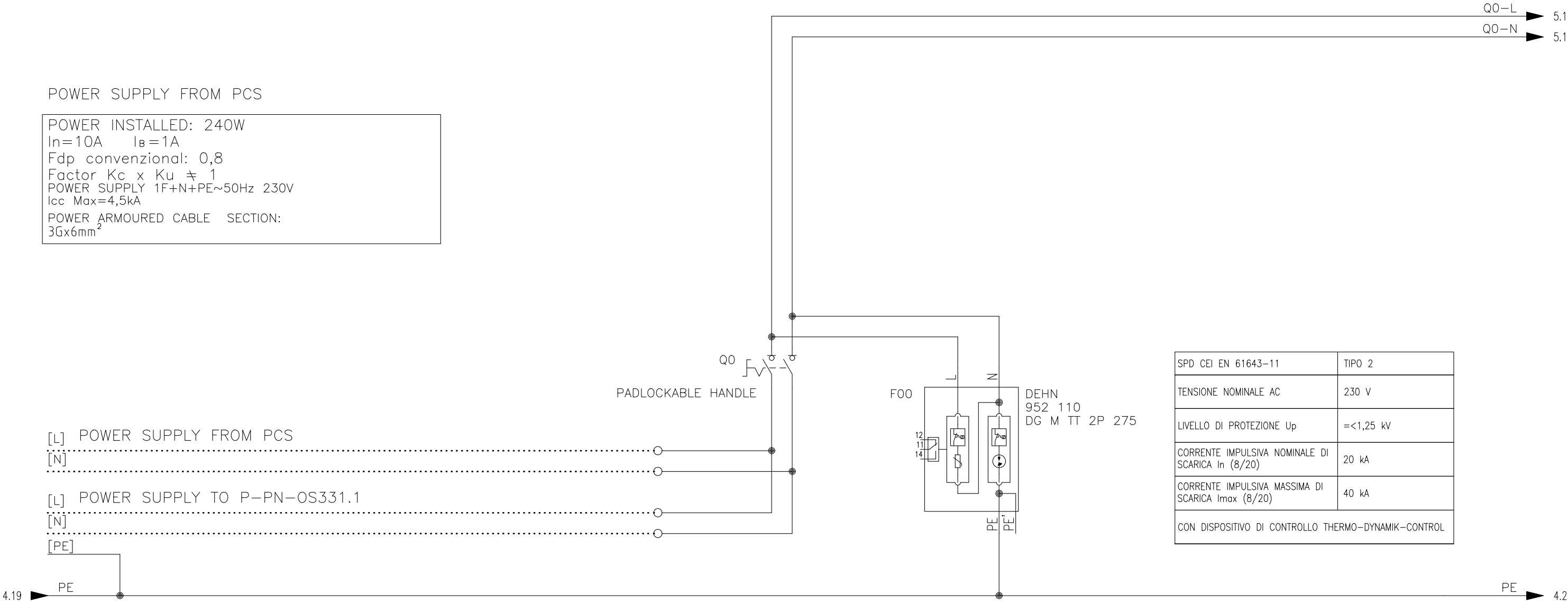
CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

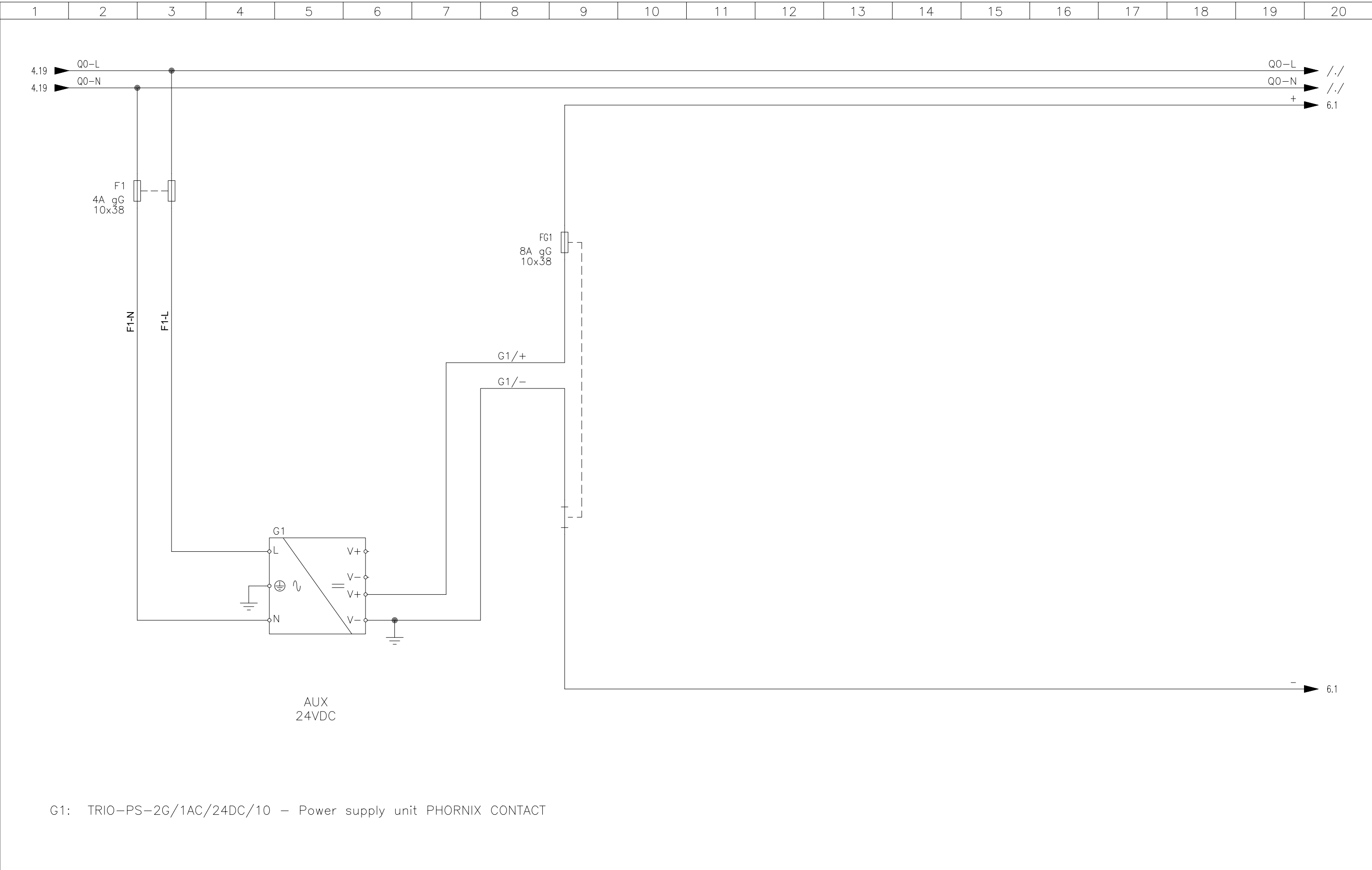
	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT–BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW–GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK–BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

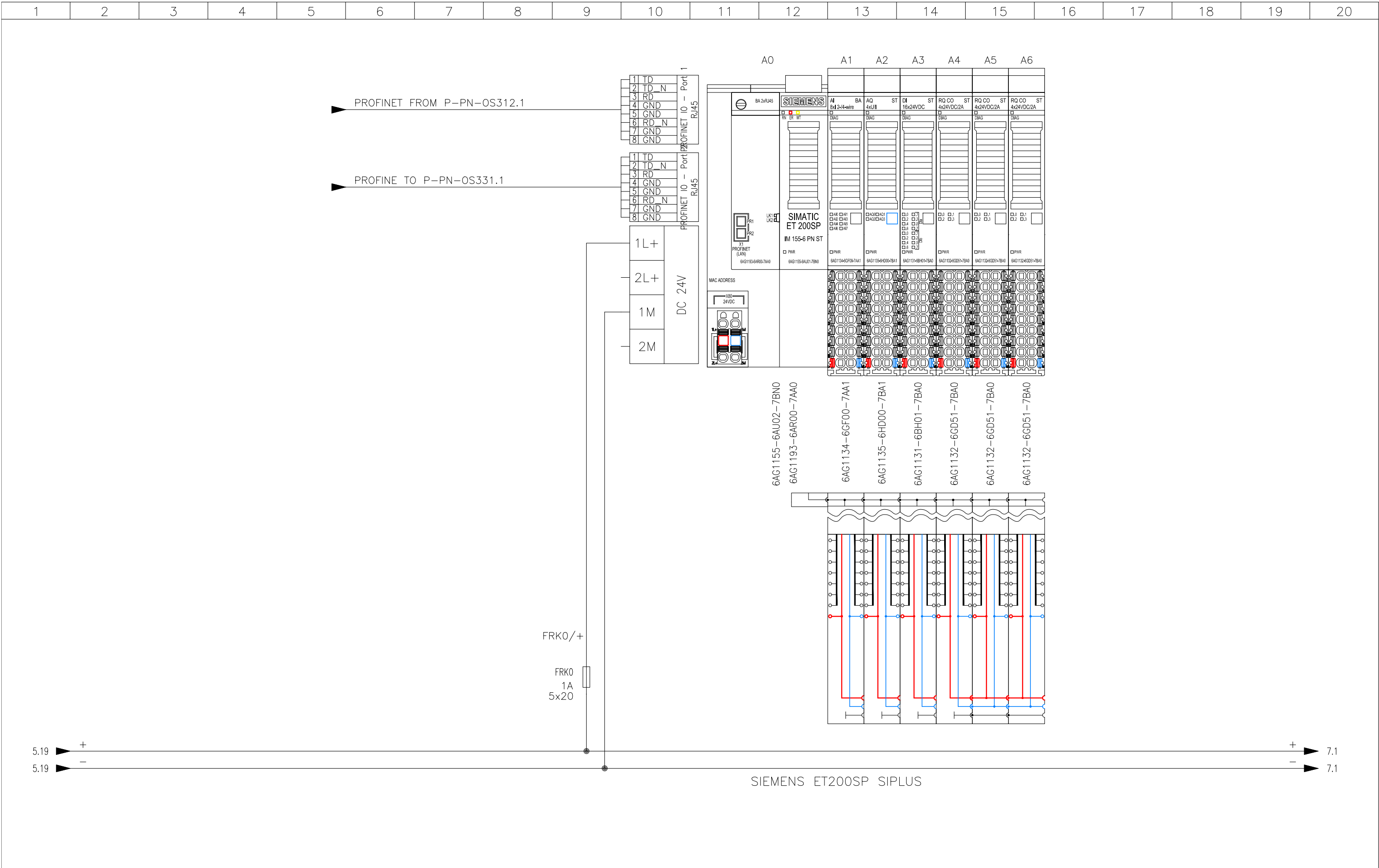
CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

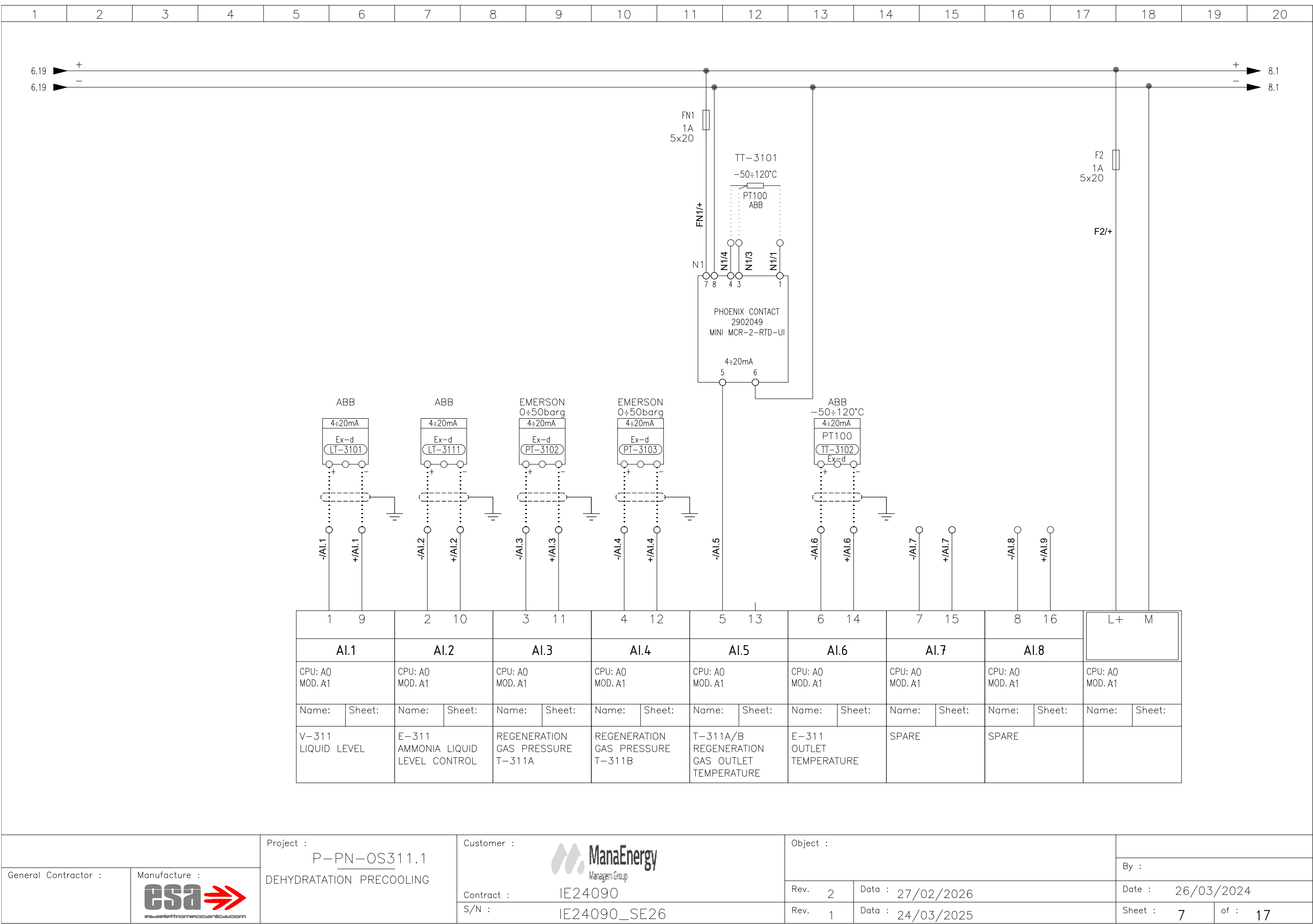
	POWER MAIN DISCONNECT SWITCH																FEEDER ID		
	1F+N+PE~50Hz 230V																RATED POWER P _n –RATED VOLTAGE V _n		
																	RATED CURRENT		

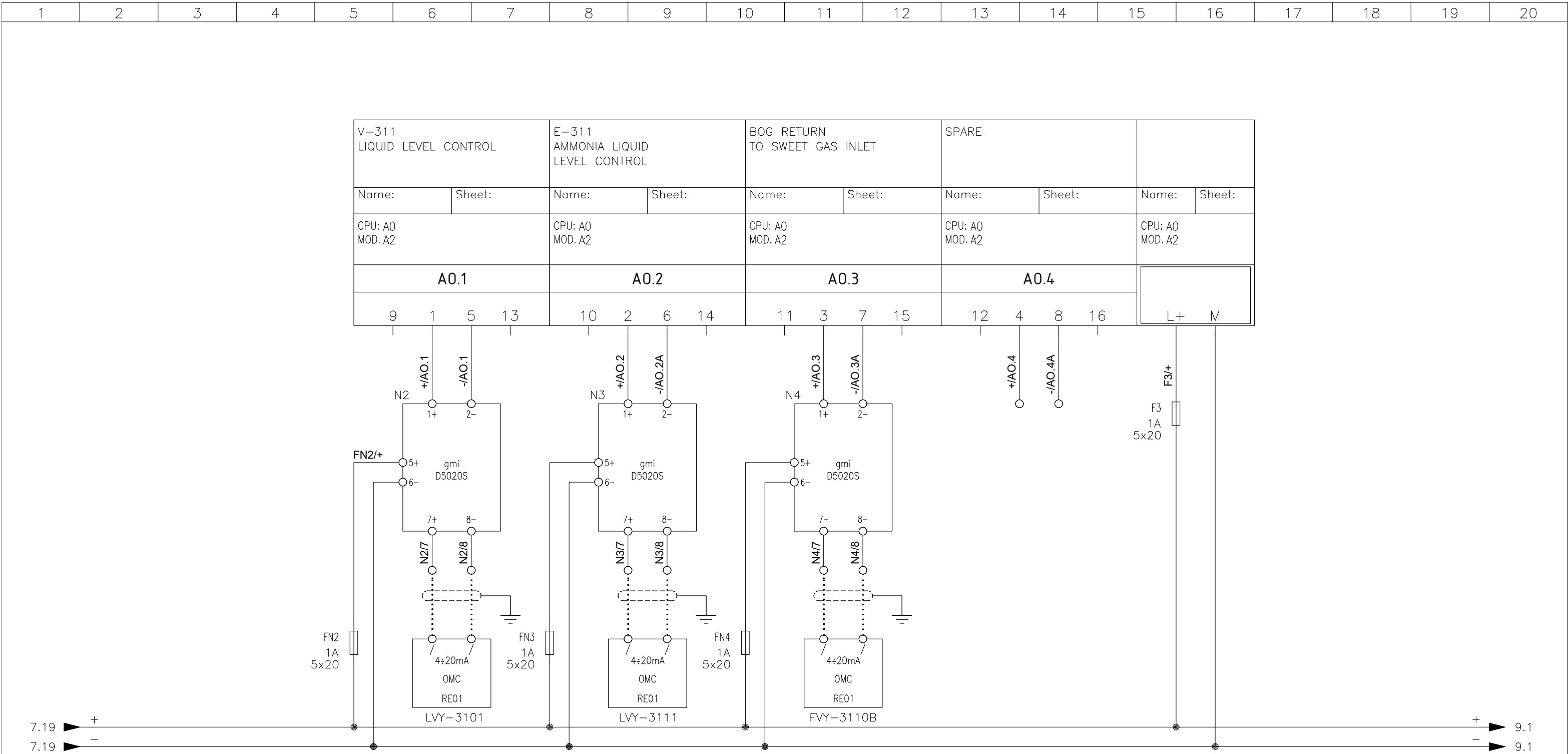


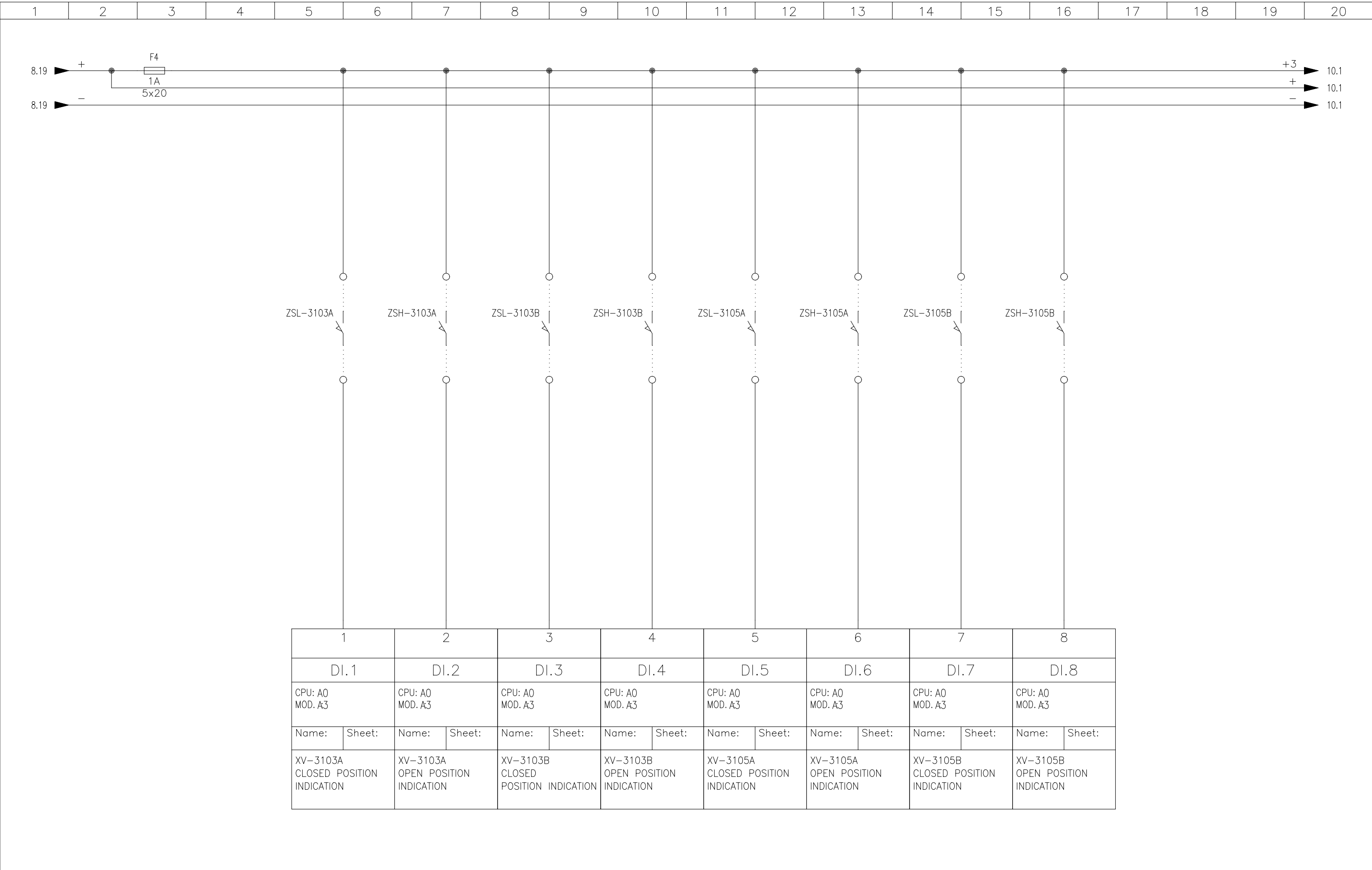


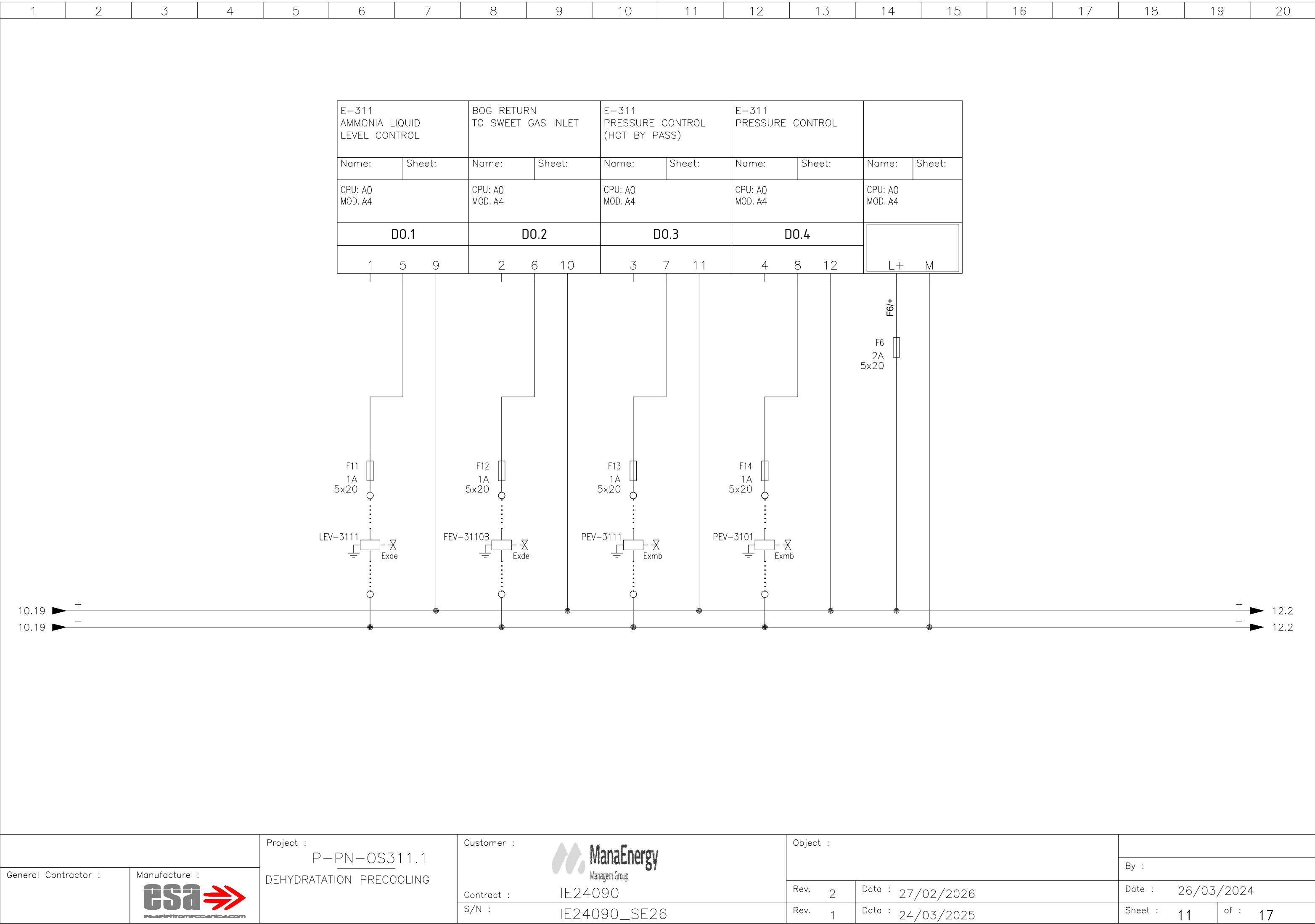
General Contractor : Manufacture : 		Project : P-PN-OS311.1 DEHYDRATATION PRECOOLING	Customer :  Contract : IE24090 S/N : IE24090_SE26		Object : Rev. 2 Data : 27/02/2026 Rev. 1 Data : 24/03/2025	By :	
						Date : 26/03/2024 Sheet : 5 of : 17	



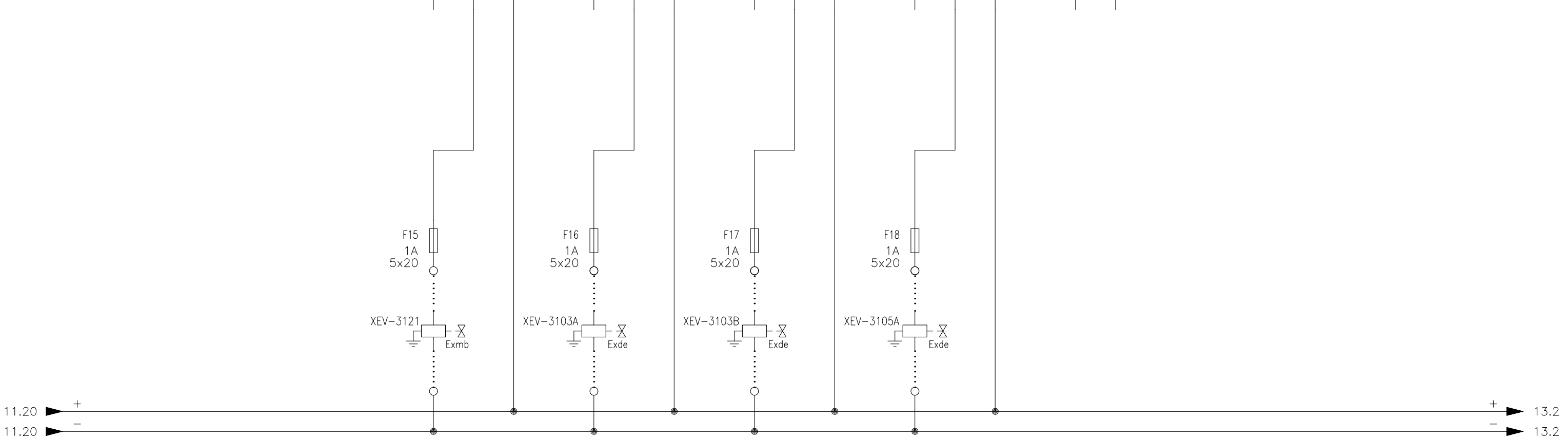


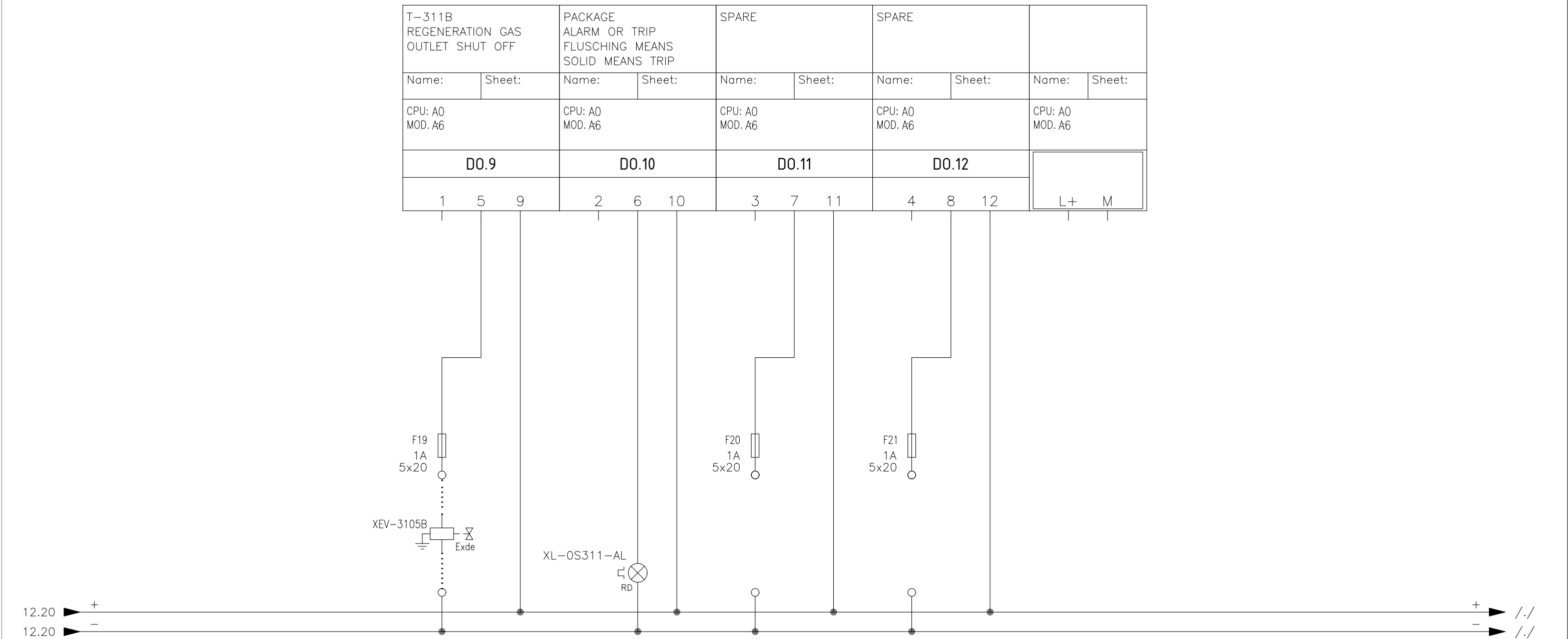


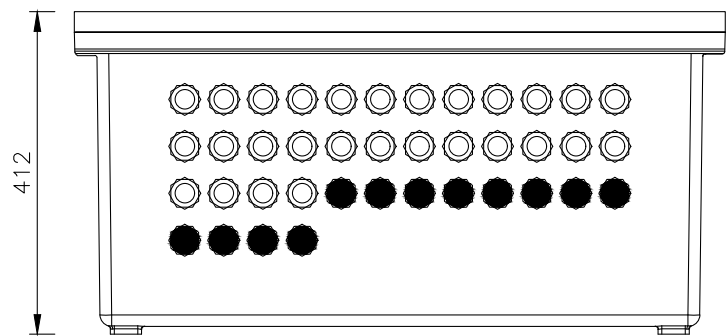
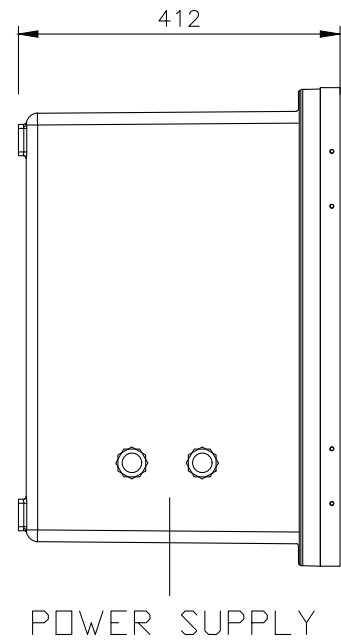
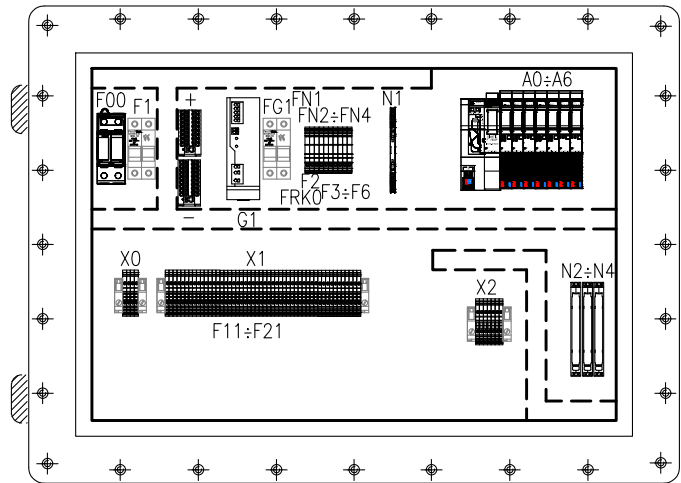
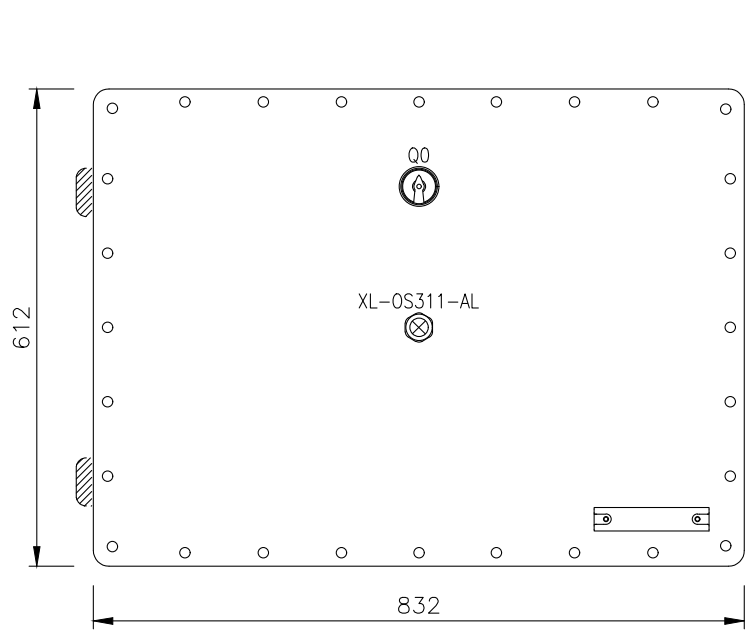




OPEN FULL BOG RECYCLE FROM REGENERATION LOOP			T-311A INLET SHUT OFF			T-311B INLET SHUT OFF			T-311A REGENERATION GAS OUTLET SHUT OFF				
Name:		Sheet:	Name:		Sheet:	Name:		Sheet:	Name:		Sheet:	Name:	Sheet:
CPU: A0 MOD. A5			CPU: A0 MOD. A5			CPU: A0 MOD. A5			CPU: A0 MOD. A5			CPU: A0 MOD. A5	
D0.5			D0.6			D0.7			D0.8			L+M	
1	5	9	2	6	10	3	7	11	4	8	12		







MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-H
230 Vac 50 Hz 16 A
Max Dis. P: 69 W
T. Cable
Year s.n 25/CMC/0183

CE 0722 Ex II 2GD
INERIS18ATEX0026X/00
Ex db [ia Ga] IIB T6
Ex tb [ia Da] IIC T85 °C IP66
-20°C < Tamb < +50°C

Cable entries: see instructions
WARNINGS: DO NOT OPEN WHILE ENERGIZED
DO NOT OPEN IF AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT
READ INSTRUCTIONS NOTE BEFORE INSTALLATION AND USE

General Contractor : Manufacture : 		Project : P-PN-OS311.1 DEHYDRATATION PRECOOLING	Customer :  Contract : IE24090 S/N : IE24090_SE26	Object :	By :	
					Rev. 2	Date : 27/02/2026
					Rev. 1	Date : 24/03/2025
					Sheet : 14	of : 17

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20				
INITIALS	SUPPLIER PN		DESCRIPTION							SUPPLIER													
–	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block							PHOENIX CONTACT													
+	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block							PHOENIX CONTACT													
A0	6AG1155–6AU01–7BN0		ET 200SP, PROFINET interface module IM 155–6PN Standard							SIEMENS													
A0	6AG1193–6AR00–7AA0		ET 200SP BA 2xRJ45							SIEMENS													
A1	6AG1134–6GF00–7AA1		AI 8xl 2–/4–wire BA							SIEMENS													
A1	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals							SIEMENS													
A2	6AG1135–6HD00–7BA1		ET 200SP AQ 4xU/I Standard							SIEMENS													
A2	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals							SIEMENS													
A3	6AG1131–6BH01–7BA0		ET 200SP DI 16x24 V DC standard							SIEMENS													
A3	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals							SIEMENS													
A4	6AG1132–6GD51–7BA0		ET 200SP RQ 4x24VDC/2A CO ST							SIEMENS													
A4	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals							SIEMENS													
A5	6AG1132–6GD51–7BA0		ET 200SP RQ 4x24VDC/2A CO ST							SIEMENS													
A5	6AG1193–6BP00–7BA0		ET 200SP BU15–P16+A0+2B push–in terminals							SIEMENS													
A6	6AG1132–6GD51–7BA0		ET 200SP RQ 4x24VDC/2A CO ST							SIEMENS													
A6	6AG1193–6BP00–7BA0		ET 200SP BU15–P16+A0+2B push–in terminals							SIEMENS													
F00	952110		DG M TT 2P 275 – SPD							DEHN													
F1	2302038		Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V							ITALWEBER													
F2	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F3	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F4	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F5	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F6	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F11	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F12	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F13	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F14	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F15	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F16	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F17	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F18	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F19	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F20	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
F21	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
FG1	2301138		Fuseholder for cylindrical fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V							ITALWEBER													
FN1	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
FN2	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
FN3	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
FN4	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
FRK0	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE													
G1	2903149		TRIO–PS–2G/1AC/24DC/10 – Power supply unit							PHOENIX CONTACT													
N1	2902049		MINI MCR–2–RTD–UI – Resistance thermometer measuring transducer							PHOENIX CONTACT													
N2	D5020S		I.S. SIL2 HART? Isolating Driver							SIEMENS													
N3	D5020S		I.S. SIL2 HART? Isolating Driver							SIEMENS													
N4	D5020S		I.S. SIL2 HART? Isolating Driver							SIEMENS													
P–PN–OS311.1	EJB–H		ENCLOSURE 832x612x412 mm L x H x D							CMC													
Q0	PSRC–320		SELECTOR SWITCH							CMC													
XL–OS311–AL	PLD20		PILOT LAMP RED							CMC													
XL–OS311–AL	020PD9		LAMPHOLDER BA9S, DIRECT SUPPLY							ELFI													
General Contractor :			Manufacture : 			Project : P–PN–OS311.1 DEHYDRATATION PRECOOLING			Customer :  Contract : IE24090 S/N : IE24090_SE26			Object :				By :							
												Rev. 2				Data : 27/02/2026				Date : 26/03/2024			
												Rev. 1				Data : 24/03/2025				Sheet : 15		of : 17	

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General Contractor :		PCS		POWER SUPPLY FROM PCS FG70AR 0,6/1kV 6mm ²		P-PN-OS331.1		POWER SUPPLY TO P-PN-OS331.1 FG70AR 0,6/1kV 6mm ²		X0		1		2		3		4		X0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
Project :		P-PN-OS311.1		Customer :				Object :																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	

INTRINSIC CIRCUIT INSIDE

